LAB 3 REPORT

CYOP 300 - BUILDING SECURE WEB APPLICATIONS

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Executive Summary

The purpose of this project was to create a simple menu in the console for the user to interact with. Each item in the menu performed a different function related to the 50 U.S. states, with options such as displaying each of the states in alphabetical order, searching for the information regarding a specific state such as the population, state capital, state flower, showing the top 5 U.S. states ordered by population, and the option to update a specific state's population information.

The menu should repeatedly ask the user for what function they want to do, and then check each of their inputs to make sure that it fits the semantics of the expected type. The program only finishes when the user says that they're finished.

All of the requirements for this assignment were fulfilled, including earning a pylint score of 10.00/10. This was done by breaking down each function into its own methods and submethods, which kept the number of statements and return statements to a minimum.

Testing

Test Case Table

Test Cases	Description	Expected Output	Actual Output	Pass/Fail
Test Case 1	Menu item 1 working as expected	All US states listed in alphabetical order	All US states listed in alphabetical order	pass
Test Case 2	Searching for a specific state by name	Input: Maryland Output: information	Input: Maryland Output: information	pass
Test Case 3	Searching for a specific state by code	Input: Md Output: information	Input: Md Output: information	pass
Test Case 4	Incorrectly entering a name for a state	Input: Texa5 Output: Error	Input: Texa5 Output: Error	pass
Test Case 5	Menu item 3 working as expected	Showing the top 5 populated states	Showing the top 5 populated states	pass
Test Case 6	Entering a state's name and updating population to 50	Input: CA, 50 Output: success	Input: CA, 50 Output: success	pass
Test Case 7	Incorrectly entering a state's name	Input: AA Output: ERROR	Input: AA Output: ERROR	pass
Test Case 8	Entering in a negative population	Input: CA, -50 Output: ERROR	Input: CA, -50 Output: ERROR	Pass
Test Case 9	Changing the #1 populated state, then checking the new top 5 state graph	Input: CA, 50 Output: Success, California is missing from graph	Input: CA, 50 Output: Success, California is missing from graph	pass
Test Case 10	Making Maryland the #1 populated state, then checking the new top 5 state graph	Input: MD, 50000000 Output: Success, Maryland is higher than california	Input: MD, 50000000 Output: Success, Maryland is higher than california	pass
Test Case 11	Menu item 5	Exits correctly	Exits correctly	pass
Test Case 12	Attempting menu item 6	ERROR	ERROR	pass

Test Case Screenshots

Test Case 1

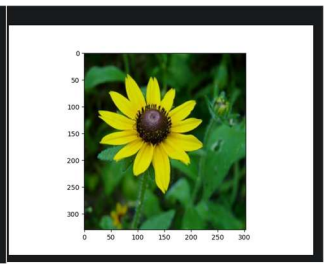
```
Choose an item from the menu below:
    1.  Display all U.S. States in Alphabetical Order
    2.  Search For a Specific State
    3.  Show Top 5 Populated States
    4.  Update a State's Population
    5.  Exit
1
NAME: Alabama, CAPITAL: Montgomery, POPULATION: 5,237,750, STATE FLOWER: Camellia
NAME: Alaska, CAPITAL: Juneau, POPULATION: 747,379, STATE FLOWER: Alpine Forget-me-not
NAME: Arizona, CAPITAL: Phoenix, POPULATION: 7,801,100, STATE FLOWER: Saguaro cactus blossom
NAME: Arkansas, CAPITAL: Little Rock, POPULATION: 3,126,140, STATE FLOWER: Apple blossom
NAME: California, CAPITAL: Sacramento, POPULATION: 39,896,400, STATE FLOWER: California poppy
NAME: Colorado, CAPITAL: Denver, POPULATION: 6,069,800, STATE FLOWER: Rocky Mountain columbine
NAME: Connecticut, CAPITAL: Hartford, POPULATION: 3,739,160, STATE FLOWER: Mountain Laurel
NAME: Delaware, CAPITAL: Dover, POPULATION: 1,082,900, STATE FLOWER: Peach Blossom
```

Test Case 2

```
Maryland
State facts for Maryland:
    Capital: Annapolis
    Population: 6,355,540
    State Flower: Black-Eyed Susan
```

Test Case 3

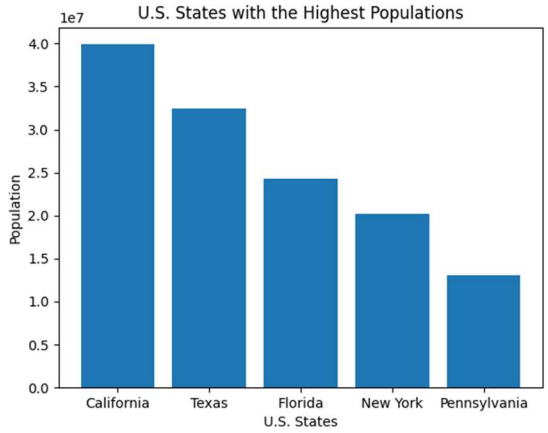
```
md
State facts for Maryland:
    Capital: Annapolis
    Population: 6,355,540
    State Flower: Black-Eyed Susan

****************************************
   Thank you for using the program!
****************************************
```



Test Case 4

```
Texa5
ERROR: State not found. Check your spelling and try again.
```

Test Case 5



Test Case 6

```
ca
State facts for California:
    Capital: Sacramento
    Population: 50
    State Flower: California poppy
```

Test Case 7

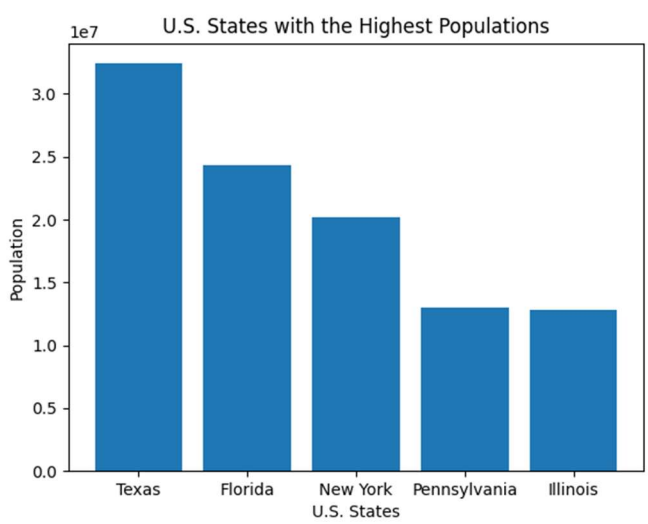
```
aa
ERROR: State not found. Check your spelling and try again.
```

Test Case 8

```
Search for a state by entering its name or its 2-character abbreviation: ca
SELECTED STATE: California
CURRENT POPULATION: 39,896,400

What would you like to set the population to?:
-50
ERROR: Please enter a positive whole number for the new population

Search for a state by entering its name or its 2-character abbreviation: ca
SELECTED STATE: California
CURRENT POPULATION: 39,896,400

What would you like to set the population to?:
50
Population set to 50
```

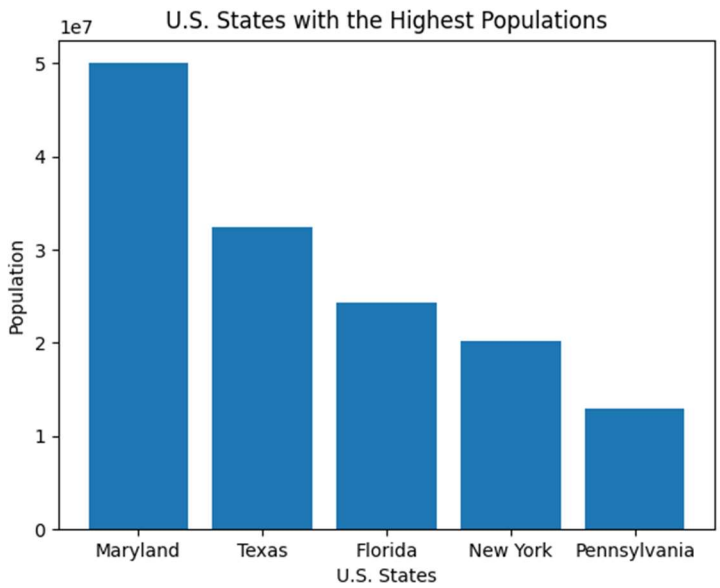
Test Case 9

```
Search for a state by entering its name or its 2-character abbreviation: ca
SELECTED STATE: California
CURRENT POPULATION: 39,896,400

What would you like to set the population to?:
50
Population set to 50
```



Test Case 10

```
Search for a state by entering its name or its 2-character abbreviation: md
SELECTED STATE: Maryland
CURRENT POPULATION: 6,355,540

What would you like to set the population to?:
50000000
Population set to 50000000
```



Test Case 11

```
Choose an item from the menu below:
    1.  Display all U.S. States in Alphabetical Order
    2.  Search For a Specific State
    3.  Show Top 5 Populated States
    4.  Update a State's Population
    5.  Exit
5

****************************************
   Thank you for using the program!
****************************************
```

Test Case 12

```
Choose an item from the menu below:
    1.  Display all U.S. States in Alphabetical Order
    2.  Search For a Specific State
    3.  Show Top 5 Populated States
    4.  Update a State's Population
    5.  Exit
6
ERROR: Please select an option 1-5
```

Pylint Results

```
PS C:\Users\Mason\OneDrive\Desktop\UMGC\CYOP 300\Lab 3> pylint main.py

------------------------------------------------------------------
Your code has been rated at 10.00/10 (previous run: 10.00/10, +0.00)
```